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Brief Communication

BRIEF COMMUNICATION

Black–White Breast Cancer Incidence Trends: Effects of Ethnicity

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Abstract

Recent reports of converging black and white breast cancer incidence rates have gained much attention, potentially foreshadowing a worsening of the black–white breast cancer mortality disparity. However, these incidence rates also reflect the sum of non-Hispanics and Hispanics that may mask important ethnicity-specific trends. We therefore assessed race- and ethnicity-specific breast cancer trends using the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) 13 Registries Database (1992–2014). Age-period-cohort models projected rates for 2015–2030. Results confirmed merging of age-standardized incidence rates for blacks and whites circa 2012, but not for non-Hispanic black (NHB) and non-Hispanic white (NHW) women. Incidence rates were highest for NHW women (n = 382 290), followed by NHB women (n = 51 074), and then Hispanic white women (n = 48 651). The sample size for Hispanic blacks was too small for analysis (n = 693). Notably, future incidence rates are expected to slowly increase (2015 through 2030) among NHW women (0.24% per year, 95% confidence interval [CI] = 0.17 to 0.32) and slowly decrease for NHB women (–0.14% per year, 95% CI = –0.15 to –0.13). A putative worsening of the black–white mortality disparity, therefore, seems unlikely. Ethnicity matters when assessing race-specific breast cancer incidence rates.

Overall breast cancer incidence rates in the United States increased until 2000, sharply declined through 2005 (1–3), then stabilized (4). Although white women have historically experienced higher breast cancer incidence than black women, black and white breast cancer rates appeared to converge circa 2012 (Figure 1A), reflecting stable rates among whites and rising rates among blacks (5,6). These racial trends were heralded as a reversal in the historically lower incidence rates among blacks, possibly even foreshadowing a worsening of the black-to-white breast cancer mortality disparity (6) and motivating new breast cancer research (7–13).

However, race-only population-based rates may mask important ethnicity-specific trends among non-Hispanics and Hispanics. We therefore analyzed current and future race- and ethnicity-specific female breast cancer trends using the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) 13 Registries Database (November 2016 submission, 1992–2014) and SEER*Stat software (www.seer.cancer.gov/seerstat). Incidence rates were age-standardized to the 2000 US standard population for women age 30 to 84 years. Future trends were projected with age-period-cohort models, as previously described (2,14).

Of 482 015 breast cancer case patients in our analysis, 382 290 were non-Hispanic white (NHW), 51 074 were non-Hispanic black (NHB), and 48 651 were Hispanic white (HW) women age 30 to 84 years, with 224 176 446 woman-years of follow-up in SEER 13 from 1992 through 2014. There were too few Hispanic black case patients for analysis (n = 693).

Figure 1A shows observed and projected invasive breast cancer incidence rates for black and white race, with rates further stratified by Hispanic ethnicity in Figure 1B. Results confirmed a merging of breast cancer incidence rates for all blacks and all whites circa 2012 (Figure 1A), a convergence that was not present for NHW and NHB women (Figure 1B). Indeed, rates were

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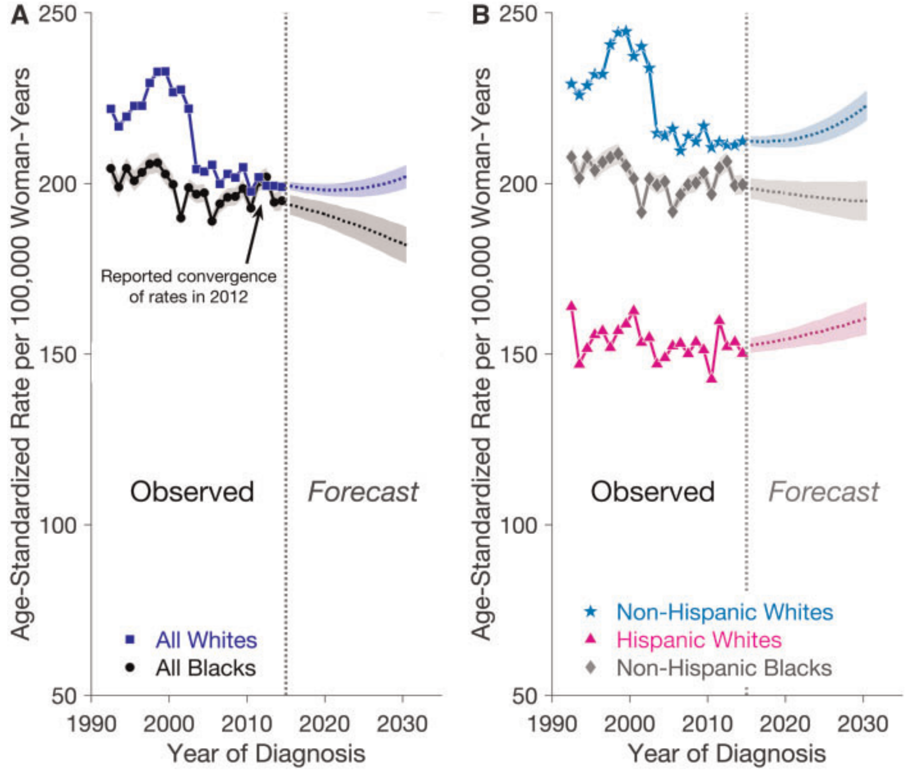

Figure 1. Observed and projected incidence of invasive female breast cancer in the Surveillance, Epidemiology, and End Results (SEER) 13 registries by race and ethnicity. **A)** Observed (1992–2014) and projected (2015–2030) age-standardized incidence rates of invasive breast cancer per 100 000 woman-years for all white (**squares**) and all black (**circles**) women between the ages of 30 and 84 years in the SEER 13 registries. An **arrow** points to the previously reported convergence in rates between all white and all black women for the year 2012 (5). **B)** Observed (1992–2014) and projected (2015–2030) age-standardized incidence rates of invasive breast cancer per 100 000 woman-years for non-Hispanic white (**stars**), non-Hispanic black (**diamonds**), and Hispanic women (**triangles**) between the ages of 30 and 84 years in the SEER 13 registries. In each panel, a **vertical reference line** separates the observed from the forecast period. **Shaded bands** show point-wise 95% confidence limits for the forecasted incidence rates, denoted to the **right of the vertical reference line.**

highest for NHW women, followed by NHB and HW women, and none of these three curves crossed. Moreover, forecasted breast cancer rates are expected to slowly widen between NHB and NHW women (2015 through 2030), with rising rates among NHW women (0.24% per year, 95% confidence interval [CI] = 0.17 to 0.32) and declining rates for NHB women (–0.14% per year, 95% CI = –0.15 to –0.13). Incidence rates are also expected to increase moderately for HW women (0.33% per year, 95% CI = 0.31 to 0.35).

Our analysis suggests that overall breast cancer incidence rates are not converging among NHB and NHW women. In fact, rates are expected to diverge, that is, to slowly increase among NHWs and slowly decrease among NHB. The potential reasons for differences in breast cancer incidence rates by racial/ethnic group are complex and may include differences in behavioral and etiologic factors between these groups that can affect incidence, such as participation in screening programs, obesity, age at first birth and number of births, breastfeeding history, or menopausal hormone therapy use (15–20). Furthermore, breast cancer incidence rates reflect the sum of incident estrogen receptor (ER)–positive and ER-negative tumors, and we have previously reported increasing ER-positive and decreasing ER-negative rates among US women (14). As ER-positive and -negative tumors are known to have distinct etiologies (21) and to vary in their distribution by race (22) and Hispanic ethnicity (23), a more thorough understanding of race- and ethnicity-specific breast cancer incidence trends will require comprehensive analyses by both age and hormone receptor status. Whenever possible, comparative breast cancer analyses should also attempt to analyze heterogeneous ethnic populations separately.

The strengths of this study include the use of race and ethnicity to better understand the breast cancer incidence trends for women in the SEER 13 registries, as well as the use of current data to predict future rates and provide estimates for what the breast cancer incidence landscape will look like through 2030. However, we recognize that our forecasts are the result of statistical modeling (14) that may not truly capture how this disease will evolve in the coming years. For example, our forecasts of incidence assume that the birth cohort effects estimated up until now for each race/ethnic group will hold in the future for both native-born persons and immigrants. Furthermore, the limited sample size of Hispanic black case patients precluded separate analyses for this group. Future studies that combine cancer incidence data with race- and ethnicity-specific population projections may provide additional insights into the future burden of breast cancer in the United States, where it is expected that the number of Hispanic women and those of two or more races will dramatically increase over the next 40 years (24).

Nevertheless, rather than an anticipated worsening of the black–white breast cancer racial mortality disparity, our analysis offers a more encouraging outlook for NHB women in the United States. Although it is promising to see an expected falling incidence trend for NHB women, breast cancer incidence rates in white women may continue to rise. How these forecasted patterns will translate into future trends in breast cancer mortality is unknown; however, our findings suggest that increasing racial disparities in mortality between NHB and NHW women are unlikely. Continued surveillance and additional work are needed to uncover factors that may contribute to the dissimilar breast cancer incidence trends by race/ethnicity to inform tailored prevention strategies for specific groups of women.

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